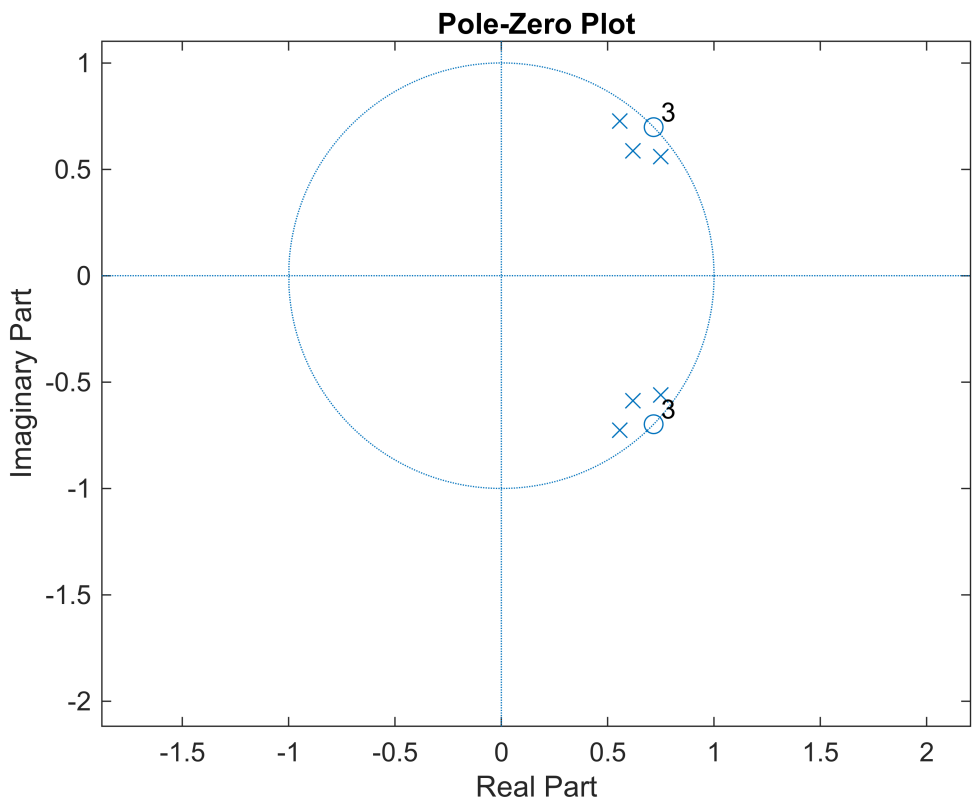
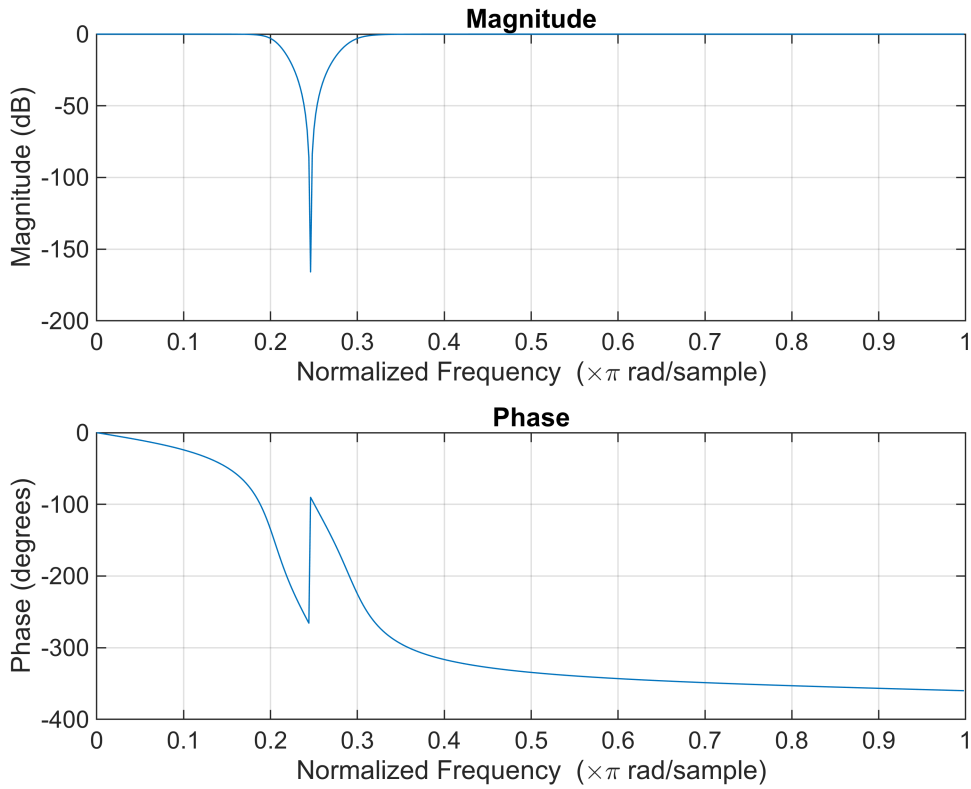
```
Fs = 8000;
f1 = 800;
f2 = 1200;
w1 = 2*f1/Fs;
w2 = 2*f2/Fs;
[b,a] = butter(3, [w1, w2], 'stop');
zplane(b,a)
```



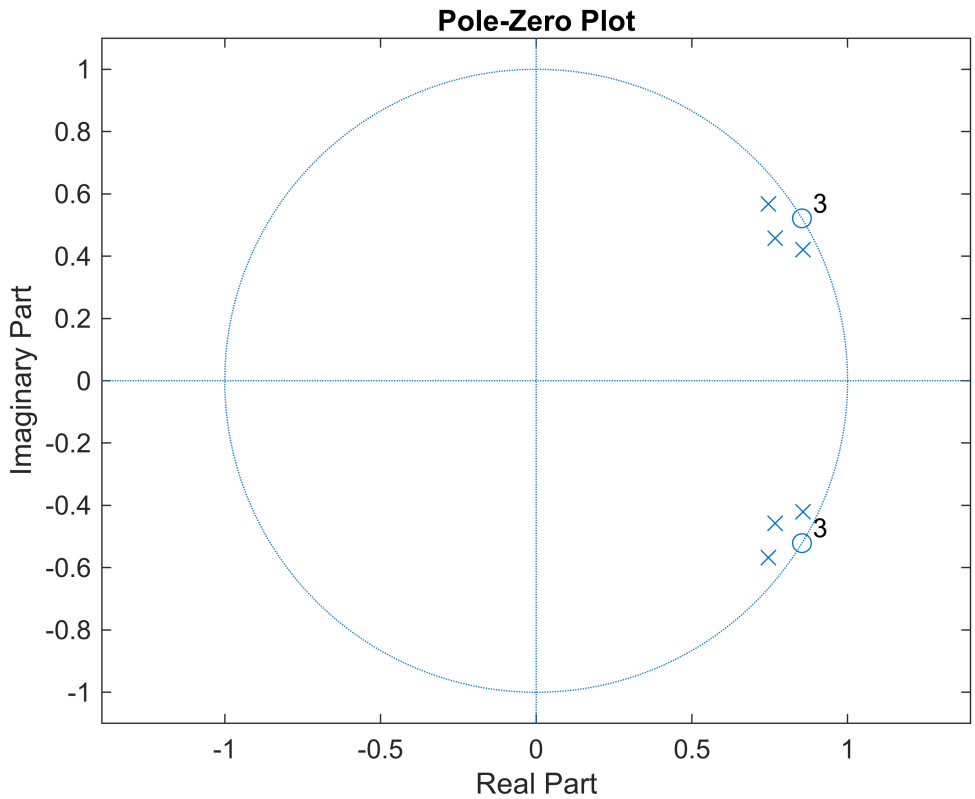
The zeros do appear where we expect them to.

```
freqz(b,a)
```

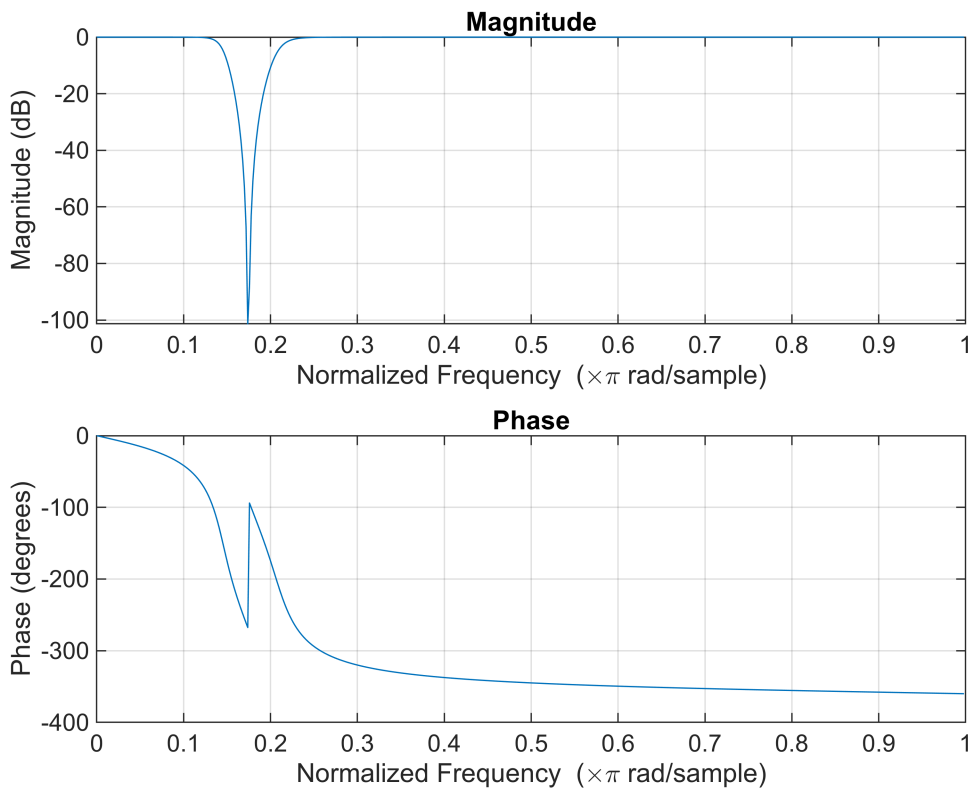


Here we see the magnitude get supressed from .2 to .3 (expected).

```
Fs = 11250;
f1 = 800;
f2 = 1200;
w1 = 2*f1/Fs;
w2 = 2*f2/Fs;
[b,a] = butter(3, [w1, w2], 'stop');

zplane(b,a)
```

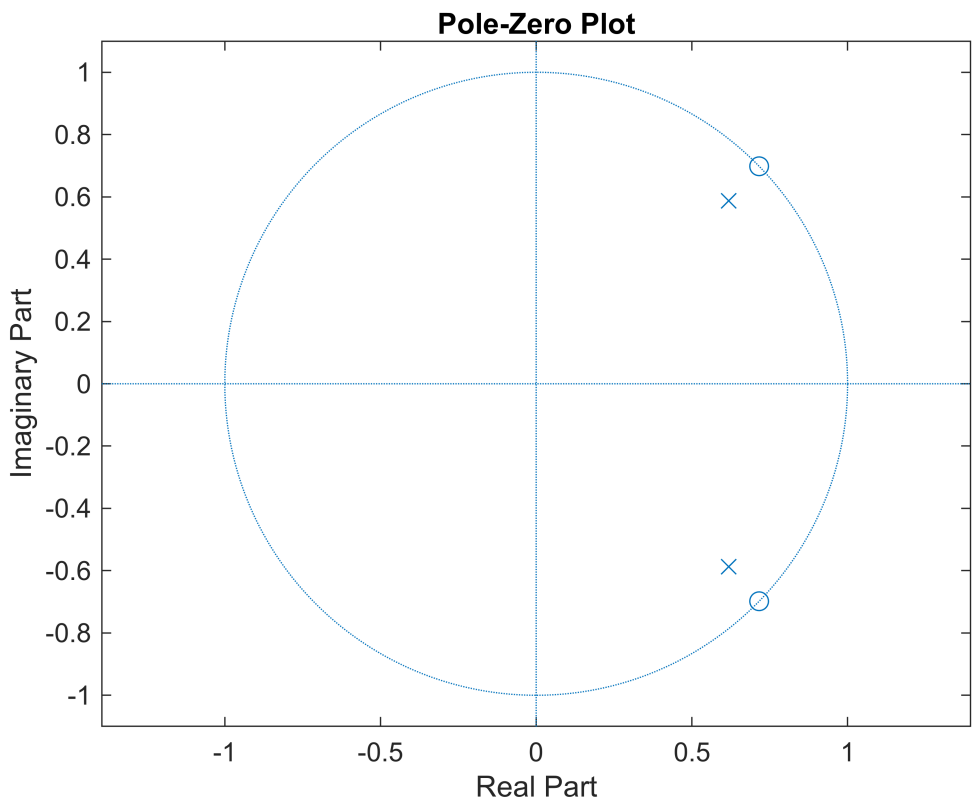

Pole-Zero Plot
Imaginary Part
Real Part
3
3

```
freqz(b,a)
```

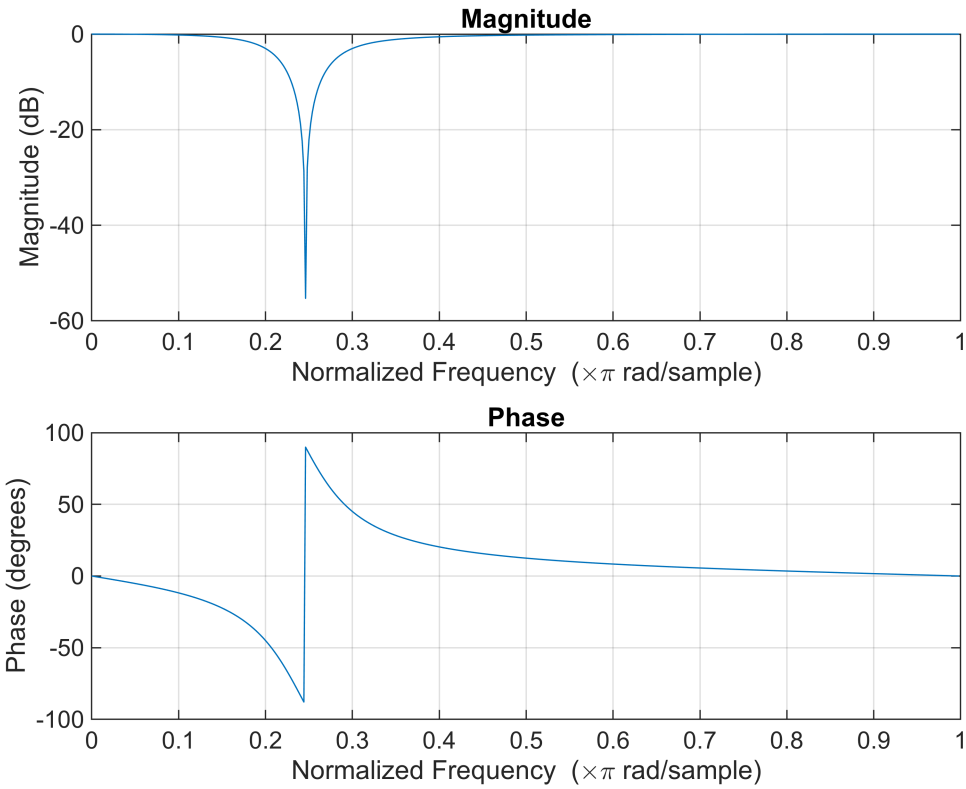

Magnitude
Magnitude (dB)
Normalized Frequency (×π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (×π rad/sample)

The zeros have shifted closer to angle = 0.

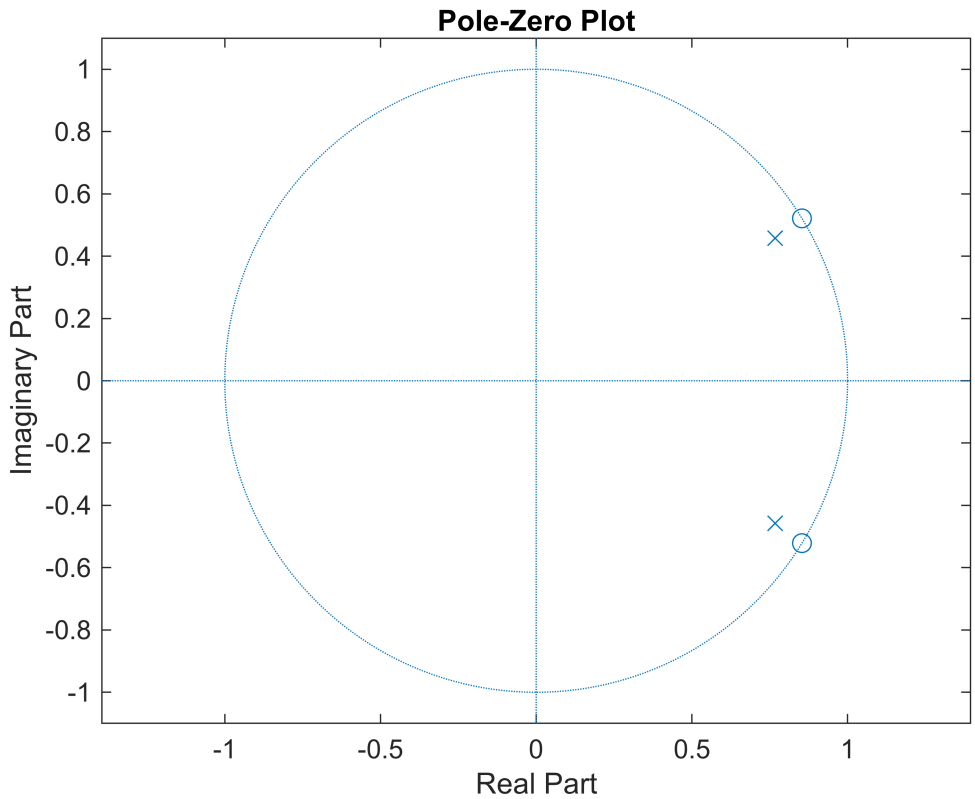
```
Fs = 8000;
f1 = 800;
f2 = 1200;
w1 = 2*f1/Fs;
w2 = 2*f2/Fs;
[b,a] = butter(1, [w1, w2], 'stop');
zplane(b,a)
```



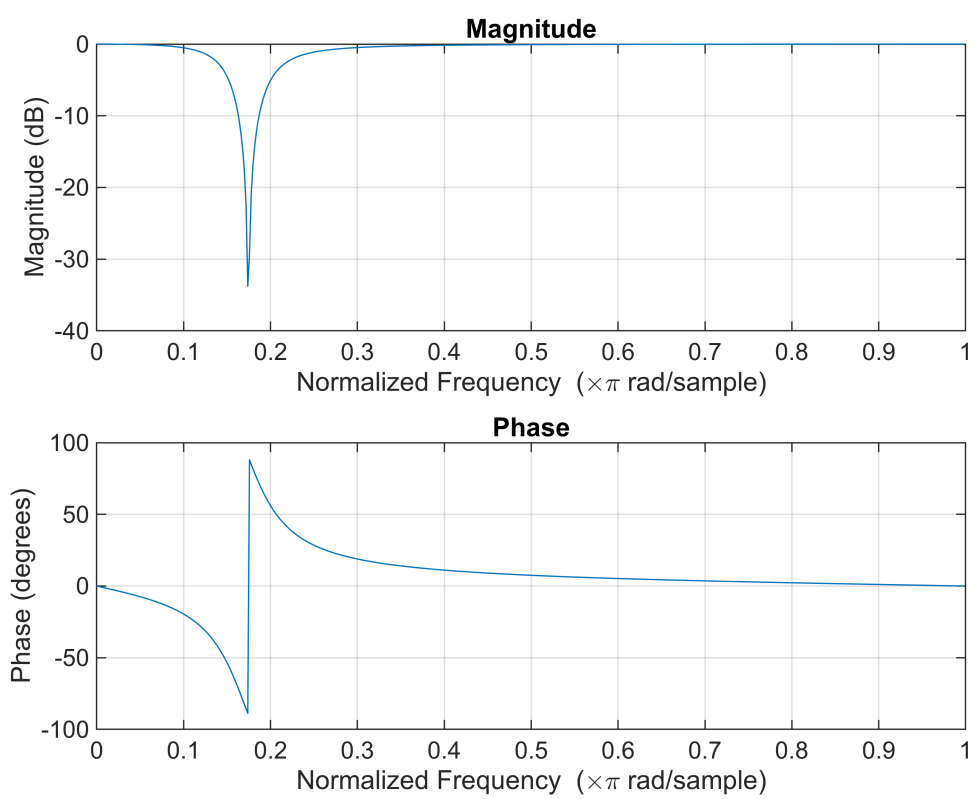

```
freqz(b,a)
```




```
Fs = 11250;
f1 = 800;
f2 = 1200;
w1 = 2*f1/Fs;
w2 = 2*f2/Fs;
[b,a] = butter(1, [w1, w2], 'stop');
zplane(b,a)
```

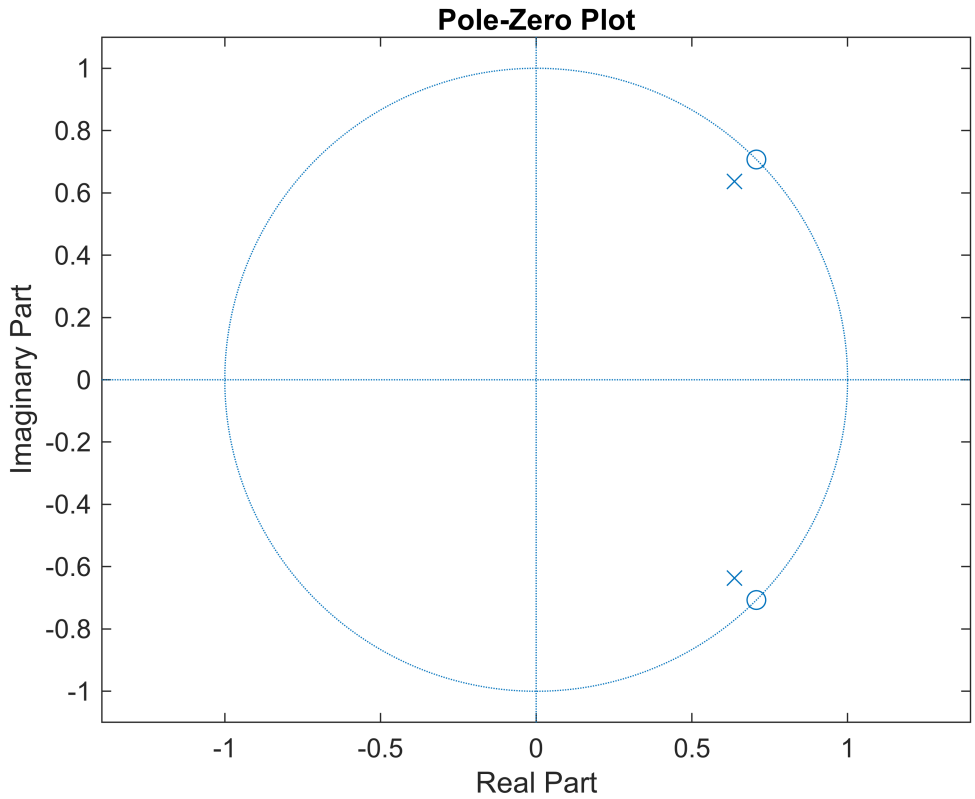

Pole-Zero Plot
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1
-0.5
0
0.5
1

```
freqz(b,a)
```

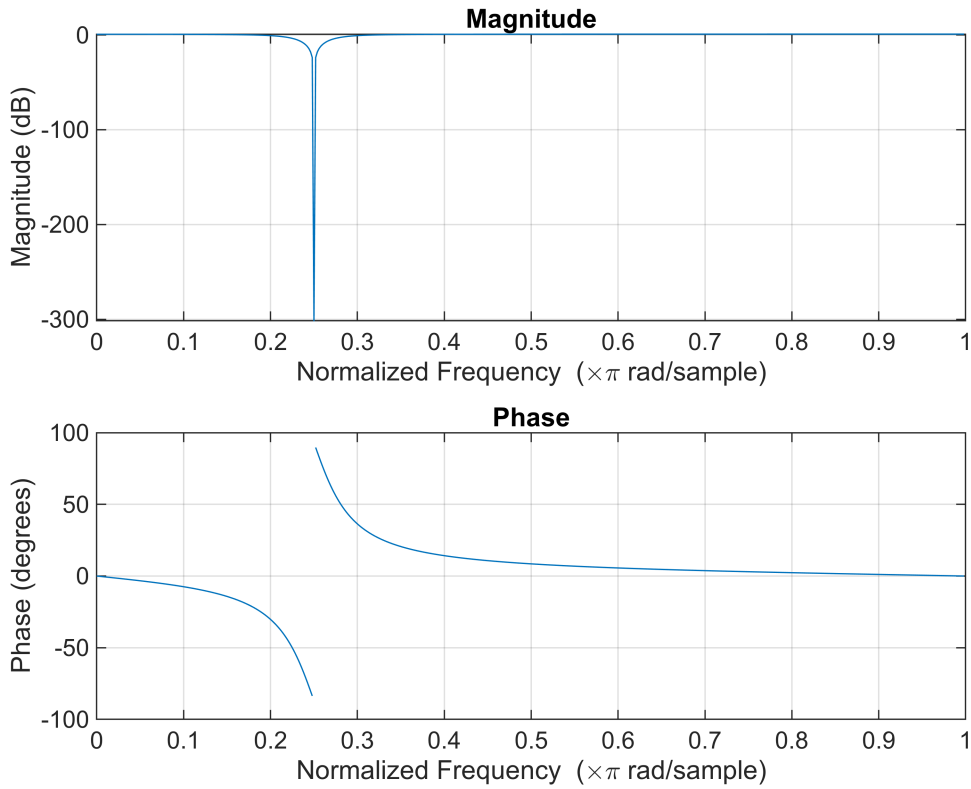

Magnitude
Magnitude (dB)
0
-10
-20
-30
-40
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (×π rad/sample)
Phase
Phase (degrees)
100
50
0
-50
-100
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (×π rad/sample)

The magnitude drop on the second order filter isn't as severe, nor is the phase difference.

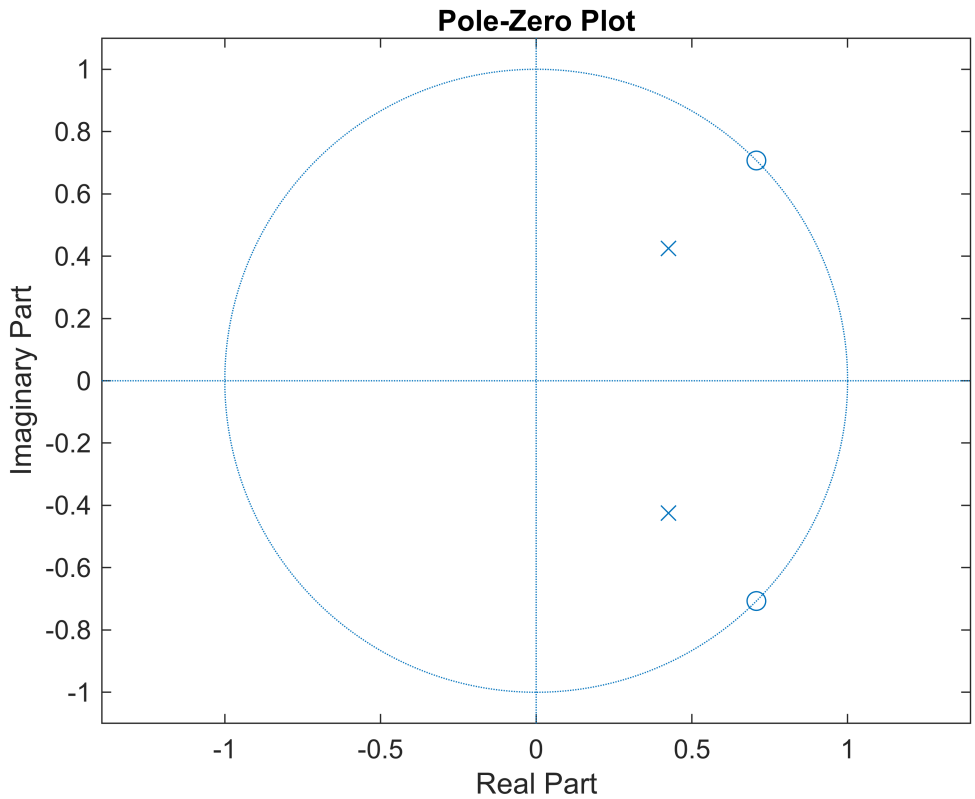
```
fo=1000;
Fs=8000;
ang=2*pi*fo/Fs;

p = 0.9;
b = [1, -2*cos(ang), 1];
a = [1, -2*p*cos(ang), p^2];
zplane(b, a)
```



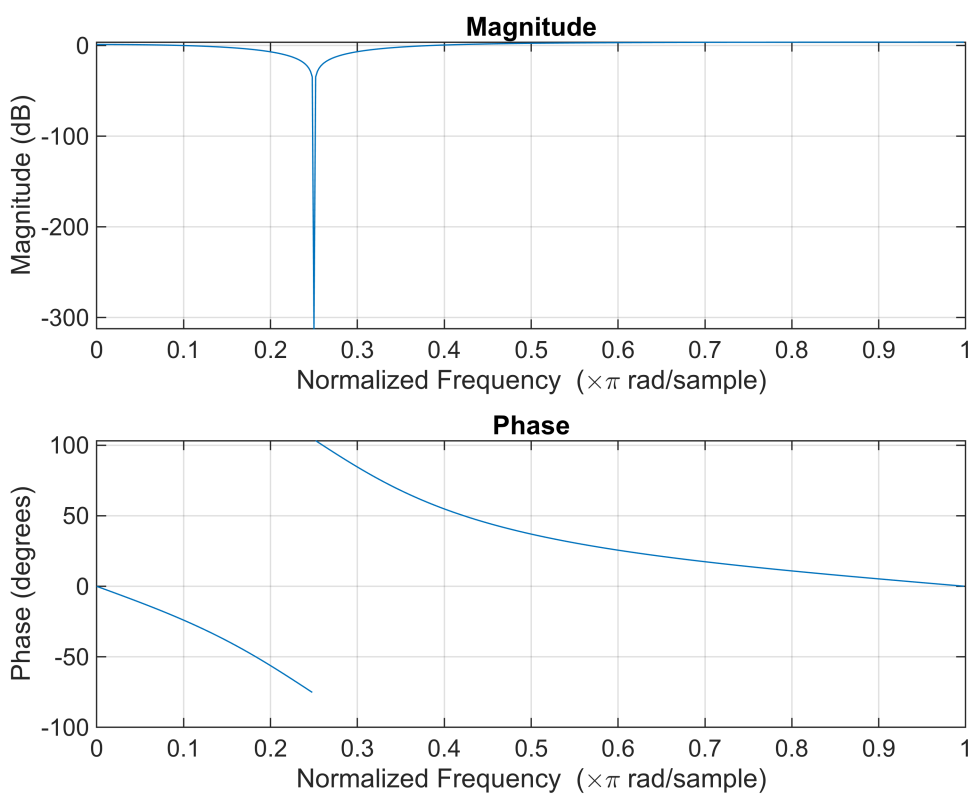

```
freqz(b, a)
```



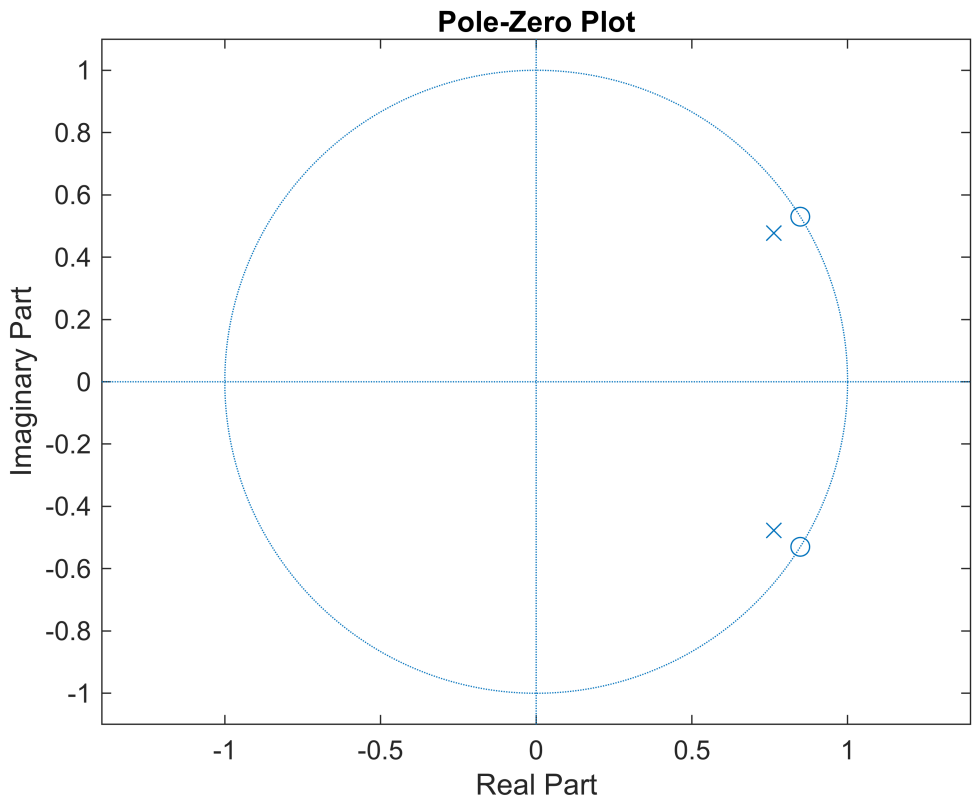

```
p = 0.6;
b = [1, -2*cos(ang), 1];
a = [1, -2*p*cos(ang), p^2];
zplane(b, a)
```


Pole-Zero Plot
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1
-0.5
0
0.5
1

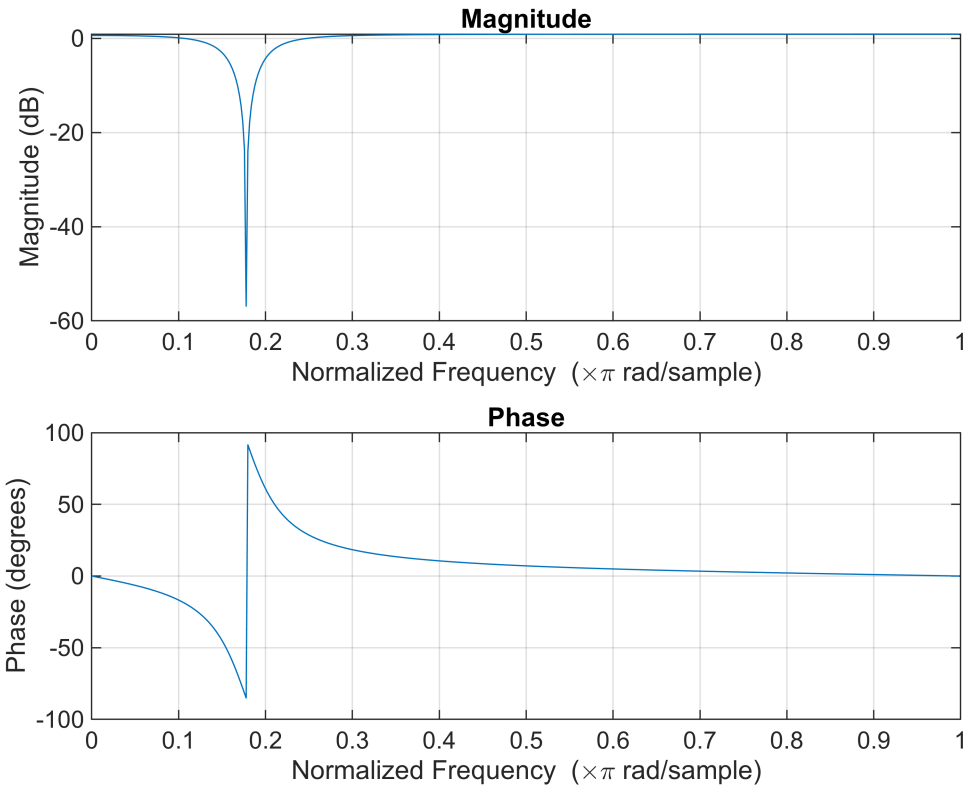
```
freqz(b, a)
```


Magnitude
Magnitude (dB)
0
-100
-200
-300
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (×π rad/sample)
Phase
Phase (degrees)
100
50
0
-50
-100
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (×π rad/sample)

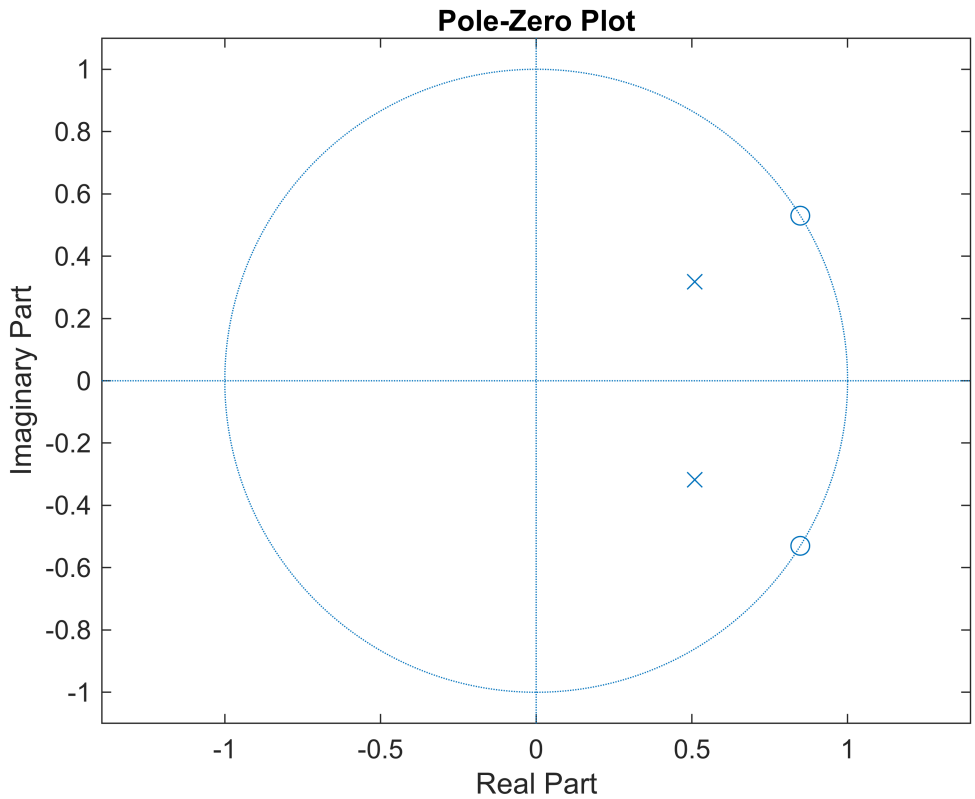
```
fo=1000;
Fs=11250;
ang=2*pi*fo/Fs;

p = 0.9;
b = [1, -2*cos(ang), 1];
a = [1, -2*p*cos(ang), p^2];
zplane(b, a)
```



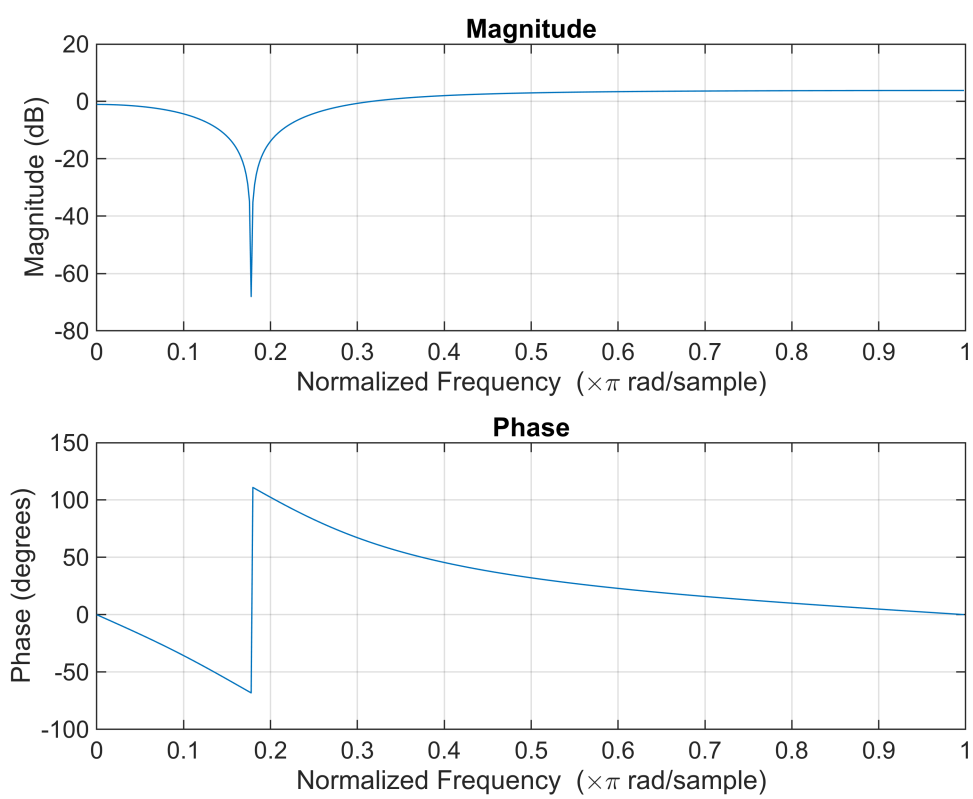

```
freqz(b, a)
```




```
p = 0.6;
b = [1, -2*cos(ang), 1];
a = [1, -2*p*cos(ang), p^2];
zplane(b, a)
```


Pole-Zero Plot
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1
-0.5
0
0.5
1

```
freqz(b, a)
```


Magnitude
Magnitude (dB)
Normalized Frequency (×π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (×π rad/sample)

The lower the p, the wider the notch. It also directly controlled how far our poles were from the origin.

Overall the notch filter provides more precise frequency rejection and the butterworth is more broad.

I would use the butterworth for a range of frequencies, and the notch to filter a singular frequency.